

ECON211: Final Exam C

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- 66 points
- 90 minutes

Probability Distributions (24 points)

Instructions: Match each of the following scenarios (1–4) with the probability distribution (A–H) that best models the random variable described (2 points per scenario). Write the letter of the distribution in the space provided. Provide an arithmetic answer to the questions (do not need to compute an exact numerical value, you can leave your answer as an arithmetic expression).

1. _____ A data scientist models the total number of fraudulent transactions X out of 500 independent credit card swipes, assuming that each swipe has a constant 0.001 probability of being a fraud.
 - (2 points) Find $P(X = 3)$.
 - (2 points) Find $Var(X)$.
2. _____ A quality control manager randomly selects 6 tablets from a bottle containing 40 tablets (10 of which are known to be slightly chipped) to check for damage. We are interested in X , the total number of chipped tablets in the sample.
 - (2 points) Find $P(X = 5)$.
 - (2 points) Find $E(X)$.
3. _____ A security technician repeatedly attempts to log into a server using a specific exploit. The probability of success is 0.05 per attempt, and each attempt is independent of others. We record the number of attempts required to achieve the first successful breach X .
 - (2 points) Find $P(X = 20)$.
 - (2 points) Find $E(X)$.
4. _____ An industrial sensor records the weight X of 1-kg flour bags coming off an assembly line. The weights vary symmetrically around the mean of 1 kg due to many small, additive, independent environmental factors.
 - (2 points) Find $P(X = 1)$.
 - (2 points) Find $P(X \geq 1)$.

Distributions:

- A. Hypergeometric Distribution
- B. Bernoulli distribution
- C. Binomial Distribution
- D. Normal Distribution
- E. Discrete Uniform Distribution
- F. Continuous Uniform Distribution
- G. Geometric Distribution
- H. Student's t -distribution.

Statistics of the Friendship Paradox (16 points)

Alice is interested in the “Friendship Paradox,” the sociological observation that, on average, a person’s friends have more friends than the person does. Alice has exactly **150 friends** on social media. She takes a random sample of **25 of her friends** and records how many friends each of them has.

The results from her sample are:

- Sample mean: $\bar{x} = 174$ friends
- Sample standard deviation: $s = 40$ friends

Alice wants to test whether her friends have, on average, *statistically significantly* more friends than she does. She sets up the following hypotheses:

$$H_0 : \mu = 150 \quad (\text{Her friends have the same number of friends as Alice}),$$

$$H_1 : \mu > 150 \quad (\text{Her friends have more friends than Alice}).$$

1. (2 points) Is this a one-sided or a two-sided test? Briefly explain why this choice is appropriate given Alice’s specific research interest.
2. (2 points) Compute the value of the t -statistic (t -score) using the sample data. Show your work.
3. At the 5% significance level ($\alpha = 0.05$), is there sufficient evidence to reject H_0 ? Provide the corresponding rejection rule based on the t -statistic and state your final conclusion.
 - (2 points) Reject H_0 if and only if t -statistic _____
 - (1 point) Conclusion (Reject or Fail to Reject): _____
4. (3 points) Suppose Alice had instead found that she **could not reject** H_0 at the 10% significance level ($\alpha = 0.10$). What would this imply? (Choose one):
 - The probability that H_0 is true is exactly 90%.
 - The p -value is greater than 0.10, meaning the observed difference could likely be due to sampling error.
 - The p -value is less than 0.10, providing evidence that Alice is exceptionally popular.
 - We have mathematically proven that μ is exactly 150.
5. (3 points) Calculate a 99% confidence interval for the true average number of friends Alice’s friends have.
6. (3 points) If the value “145 friends” does **not** belong to the 90% confidence interval, what does that mean? (Choose one):
 - There is a 10% chance that 145 is the true population mean.
 - 145 is a biased estimator of the population mean with 90% confidence.
 - If 145 were the true mean, obtaining a sample mean as extreme as 174 would have a probability of less than 10%.
 - There is a 90% probability that any given friend of Alice has at least 145 friends.

Dice (26 points)

A player's winnings in a game are determined by the sum of outcomes from rolling several independent, fair **eight-sided dice**. Let X_i be the score of the i -th die.

1. Suppose you roll only **one** die ($n = 1$).
 - (2 points) Calculate the expected value of your winnings.

 - (3 points) Calculate the variance of your winnings.

2. Now suppose you roll **two** independent dice ($n = 2$), and your total winnings $S_2 = X_1 + X_2$ is the sum of the scores.
 - (3 points) What is the probability that your total winnings are exactly 7?

 - (3 points) Calculate the expected value of your winnings.

3. Suppose you roll **20** independent dice ($n = 20$). Let S_{20} be the sum of these 20 dice. Recall that from the Central Limit Theorem (CLT), we know that the distribution of S_{20} is approximately Normal.
 - (3 points) Find the mean of S_{20} .

 - (3 points) Find the variance of S_{20} .

 - (3 points) Using CLT, calculate the approximate probability that your total winnings are less than 60. You should leave your answer in terms of the standard normal CDF, $\Phi(z)$, that is you must provide the value of z .

4. (6 points - HARD) Suppose you continue rolling a single die until the first time the side with "6" appears. Note that when the "6" appears, the game ends and you win 6 plus whatever you have made in the non-final rolls. For example, if I roll 3, 1, 6, my winnings are $3 + 1 + 6 = 10$.
What is the expected value of your winnings? Hint: use the idea of recursive expectation (or first-step analysis). You must **provide the entire solution** here.